



UNITED STATES
NUCLEAR REGULATORY COMMISSION
REGION III
2056 WESTINGS AVENUE, SUITE 400
NAPERVILLE, IL 60563-2657

November 12, 2024

Mike Mlynarek
Site Vice President
Holtec Decommissioning
International, LLC
Palisades Nuclear Plant
27780 Blue Star Memorial Highway
Covert, MI 49043-9530

SUBJECT: PALISADES NUCLEAR PLANT – RESTART INSPECTION REPORT
05000255/2024012

Dear Mike Mlynarek:

On September 30, 2024, the U.S. Nuclear Regulatory Commission (NRC) completed inspections at the Palisades Nuclear Plant. The enclosed inspection report documents these inspection results, which the inspectors discussed on November 7, 2024, with you and other members of your staff.

One Severity Level IV violation is documented in this report. We are treating this violation as a non-cited violation (NCV) consistent with Section 2.3.2 of the Enforcement Policy.

If you contest the violation or severity of the violation documented in this inspection report, you should provide a response within 30 days of the date of this inspection report, with the basis for your denial, to the U.S. Nuclear Regulatory Commission, ATTN: Document Control Desk, Washington, DC 20555-0001; with copies to the Regional Administrator, HQ; the Director, Office of Enforcement; and the NRC Lead, Palisades Restart Team.

On June 13, 2022, Palisades ceased permanent power operations and subsequently removed all fuel from the reactor, as detailed in the letter from Entergy to the NRC, "Certifications of Permanent Cessation of Power Operations and Permanent Removal of Fuel from the Reactor Vessel," (ADAMS Accession No. [ML22164A067](#)). On September 28, 2023, Holtec Decommissioning International, LLC (Holtec) submitted a letter to the NRC requesting exemptions from certain portions of the requirements of Title 10 of the *Code of Federal Regulations* (10 CFR) 50.82(a)(2) to pursue the potential reauthorization of power operations at Palisades (ADAMS Accession No. [ML23271A140](#)). The NRC's initial acceptance to review Palisades' request for exemptions was documented in ADAMS at Accession No. [ML23291A440](#) on November 3, 2023.

This letter, its enclosure, and your response (if any) will be made available for public inspection and copying at <http://www.nrc.gov/reading-rm/adams.html> and at the NRC Public Document Room in accordance with Title 10 of the *Code of Federal Regulations* 2.390, "Public Inspections, Exemptions, Requests for Withholding."

Sincerely,

A handwritten signature in cursive script that reads "April M. Nguyen".

Signed by Nguyen, April
on 11/12/24

April M. Nguyen, Lead
Palisades Restart Team
Division of Operating Reactor Safety

Docket No. 05000255
License No. DPR-20

Enclosure:
As stated

cc w/ encl: Distribution via LISTSERV®

Letter to Mike Mlynarek from April Nguyen dated November 12, 2024.

SUBJECT: PALISADES NUCLEAR PLANT – RESTART INSPECTION REPORT
05000255/2024012

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U.S. NUCLEAR REGULATORY COMMISSION
Inspection Report

Docket Number: 05000255

License Number: DPR-20

Report Number: 05000255/2024012

Licensee: Holtec Decommissioning International, LLC

Facility: Palisades Nuclear Plant

Location: Covert, MI

Inspection Dates: July 01, 2024 to September 30, 2024

Inspectors: B. Bergeon, Senior Operations Engineer
E. Fernandez, Senior Reactor Inspector
A. Hobbs, Reactor Inspector (observer)
V. Myers, Senior Health Physicist
J. Park, Reactor Inspector
A. Shaikh, Senior Reactor Inspector
J. Winslow, Senior Project Engineer

Approved By: April M. Nguyen, Lead
Palisades Restart Team
Division of Operating Reactor Safety

Enclosure

SUMMARY

This report covers announced inspections completed by regional inspectors.

The U.S. Nuclear Regulatory Commission (NRC) performed inspections of the licensee's activities to ensure the adequate protection of workers and the public under the radiation safety cornerstone and to assess requalification training and evaluation of previously licensed operators under the reactor safety cornerstones.

List of Findings and Violations

Locked High Radiation Area Identified During Chemical Decontamination			
Cornerstone	Severity	Cross-Cutting Aspect	Report Section
Not Applicable	Severity Level IV NCV 05000255/2024012-01 Open/Closed	Not Applicable	71124.01
A self-revealed Severity Level IV Non-Cited Violation (NCV) of Technical Specification (TS) 5.7.2 was identified when the licensee failed to post and control an area meeting Locked High Radiation Area (LHRA) conditions.			

Additional Tracking Items

None.

PLANT STATUS

The Palisades Nuclear Plant ceased power operations and subsequently removed all fuel from the reactor, as certified in a letter to the NRC on June 13, 2022. On September 28, 2023, Holtec Decommissioning International, LLC (Holtec) submitted a letter to the NRC requesting exemptions from certain portions of the requirements of 10 CFR 50.82(a)(2) to pursue the potential reauthorization of power operations. The NRC is conducting inspections to evaluate and assess the licensee's activities for potential restoration to an operational status.

INSPECTION SCOPES

Inspections were conducted using the appropriate portions of the referenced inspection procedures (IPs) in effect at the beginning of the inspection unless otherwise noted. Currently approved IPs with their attached revision histories are located on the public website at <http://www.nrc.gov/reading-rm/doc-collections/insp-manual/inspection-procedure/index.html>.

REACTOR SAFETY CORNERSTONES

71111.08P - Inservice Inspection Activities (PWR)

Reactor Pressure Vessel (03.01)

From July 24, 2024, to August 20, 2024, the inspectors observed and verified Holtec Palisades' nondestructive examination (NDE) activities of the Reactor Pressure Vessel conducted in accordance with the requirements of the ASME Code Section XI and MRP-227, "Materials Reliability Program: Pressurized Water Reactor Internals Inspection and Evaluation Guidelines," Revision 1-A. The purpose of these inspections was to verify that the reactor coolant system boundary, reactor vessel internals, and risk-significant piping system boundaries are appropriately monitored for degradation. The NRC inspectors also verified that the examination results were thoroughly reviewed and appropriately evaluated, and that required corrective actions to address identified indications and defects will be appropriately implemented. During this quarter, Holtec Palisades completed the NDE activities and evaluation of the data. Further NRC inspections will occur when corrective actions are implemented.

The inspectors verified that the following nondestructive examination activities were conducted appropriately per the ASME Code or required standard, and that any potential indications and defects were identified and evaluated at the proper thresholds:

MRP 227 Reactor Internals Exams (July 29, 2024, through August 12, 2024)

- Ultrasonic Testing (UT) of Core Shroud Bolts
- Visual Examination (VT) of Core Shroud Plates, Core Support Plate and Fuel Alignment Plate

ASME Section XI 10 Year ISI (August 3, 2024, through August 19, 2024)

- UT of vessel welds, vessel-to-nozzle welds and nozzle-to-pipe welds
- Phased Array Ultrasonic Testing (PAUT) of Clevis Bolts
- Visual examination (VT) of the interior surface of the vessel and supports

Steam Generator Activities (03.04)

From August 26, 2024, to September 6, 2024, the inspectors observed and verified Holtec Palisades' Steam Generator (SG) inspection activities conducted as described in the operating plant Technical Specification Requirements and EPRI Steam Generator Management Program guidance. Of note, the operating plant Technical Specifications are currently under review by the NRC. If approved, these will be implemented with the operating plant licensing basis documents if the plant goes back to an operational status. Using this as guidance for inspections now will ensure that all Technical Specification requirements will be met prior to resumption of power operations. The purpose of these inspections is to verify that the SG tube integrity is appropriately monitored and any potential mechanisms that could lead to primary-to-secondary tube leakages are addressed. The NRC inspectors verified that the inspections were properly conducted per the regulatory requirements and guidance documents, that the inspection results were thoroughly reviewed and appropriately evaluated, and that required corrective actions to address identified indications were appropriately implemented. The inspectors also reviewed Holtec Palisades' loose parts inspections, foreign material controls, and chemistry controls for the SGs. During this quarter, Holtec Palisades completed the inspection activities and were in the process of evaluating the data. Further NRC inspections will review the data analysis and any corrective actions as they are implemented.

The inspectors verified that the following inspection and cleanliness activities were conducted appropriately, and that any potential indications were identified and evaluated at the proper thresholds:

- Eddy Current Testing (ET) of 100% of Steam Generator Tubes in SG "A"
- Eddy Current Testing (ET) of 100% of Steam Generator Tubes in SG "B"
- Visual Examination (VT) of Secondary Side of SG "A" and SG "B"
- Sludge Lancing of SG "A" and SG "B"
- Foreign Objects Search and Retrieval (FOSAR) for SG "A" and SG "B"

Preliminary results of the Steam Generator inspections identified a large number of SG tubes with indications that required further analysis and/or repair. This information is documented in Preliminary Notification, PNO-III-24-002, PRELIMINARY RESULTS OF STEAM GENERATOR INSPECTIONS AT PALISADES NUCLEAR PLANT, dated September 18, 2024, (ADAMS Accession No. [ML24262A092](#)). Also, preliminary results were discussed between Holtec and NRC staff in a conference call held on September 3, 2024. This call is documented in the letter, PALISADES NUCLEAR PLANT - SUMMARY OF CONFERENCE CALL REGARDING STEAM GENERATOR TUBE INSPECTIONS, dated October 1, 2024 (ADAMS Accession No. [ML24267A296](#)).

71111.11Q - Licensed Operator Regualification Program and Licensed Operator Performance

Licensed Operator Regualification Training/Examinations (03.02)

On September 17, 2024, an NRC inspector conducted an inspection of Holtec Palisades' licensed operator requalification training program. This inspection reviewed the conduct of requalification training activities in the main control room simulator and assessed the Palisades training staff's ability to evaluate the performance of their licensed operators. The NRC inspector observed the licensed operator crew's performance during a complex casualty graded scenario and the training staff's post-scenario assessment of the crew.

No issues of concern were identified during the inspector's observation of the requalification training activities. Overall, operations training effectively evaluated the operating crew and provided effective feedback to continue to improve the performance of the operating crew.

71111.18 – Plant Modifications

Temporary Modifications and/or Permanent Modifications (03.01)

From June 17, 2024, to July 12, 2024, the inspectors evaluated temporary modifications to the primary coolant system (PCS) and the auxiliary building, 649' elevation, in support of the PCS decontamination project. The purpose of this inspection was to verify that the temporary modifications did not degrade or adversely affect the safety functions or structural integrity of the PCS and auxiliary building structure. The inspectors reviewed drawings, calculations, and evaluations of the temporary system used for the decontamination activity; performed walkdowns of the system once installed to verify appropriate installation; and observed system testing to verify no primary system leakage or adverse impacts to the normal operation of the PCS. The inspectors also verified the PCS was returned to its normal configuration after the temporary system was uninstalled.

No issues of concern were identified during the inspectors' review of the temporary modifications for this project.

RADIATION SAFETY CORNERSTONE

71124.01 - Radiological Hazard Assessment and Exposure Controls

From July 1, 2024, to July 11, 2024, the inspectors observed and evaluated Holtec Palisades' radiological assessment and controls for the PCS decontamination project. The purpose of these inspections was to evaluate the station's radiological hazards assessments, instructions to radiological workers, radiation protection access and controls to the areas affected by the project, and coverage for workers performing in-field activities for the project. The inspectors verified that the radiation protection program was appropriately implemented for the project as required by Technical Specifications, regulations, and station procedures.

The PCS decontamination activity was a process that chemically shocked the primary system piping to release radiologically contaminated build-up on the inside of the piping. Once released from the piping, the contaminated water was then sent through a series of resin beds to remove the contamination. This process was being conducted to reduce the overall contamination and radiation hazards in the system and to reduce worker dose for future activities. This was a first-time evolution for this process at Palisades.

Prior to initiation of the chemical decontamination process, the inspectors reviewed documents related to planning and control of the evolution. This included items such as system schematics, system operating procedures, planned radiological survey points, shielding packages, radiological posting plans, and radiation work permits and As-Low-As-Reasonably-Achievable (ALARA) plans. In addition, the inspectors performed several walkdowns of the system, attended pre-job briefings, and observed radiation protection staff performing posting and control changes to potentially affected areas of the plant.

During the chemical injection and subsequent clean-up phases, the inspectors continued to provide oversight of the evolution. This included various walkdowns to validate radiological postings and controls, observations of dose rates being collected throughout the site via remote monitoring capabilities, and observations of radiation protection technician job coverage.

One SLIV NCV was identified during these inspection activities, which is described in detail in the inspection results section below. Additionally, the inspectors shared an observation of note with the Palisades team during the exit meeting. This is documented in more detail in the results section below.

INSPECTION RESULTS

Locked High Radiation Area Identified During Chemical Decontamination			
Cornerstone	Severity	Cross-Cutting Aspect	Report Section
Not Applicable	Severity Level IV NCV 05000255/2024012-01 Open/Closed	Not Applicable	71124.01
A self-revealed Severity Level IV Non-Cited Violation (NCV) of Technical Specification (TS) 5.7.2 was identified when the licensee failed to post and control an area meeting Locked High Radiation Area (LHRA) conditions.			
<u>Description:</u> On July 6, 2024, Holtec Palisades initiated recirculation of clean water through the primary coolant system (PCS) chemical decontamination skid to test for leaks prior to commencing chemical injection. During this time, radiological surveys conducted by a radiation protection technician in the auxiliary building 649' elevation identified a hot spot on a piece of skid equipment that resulted in dose rates of 2 rem/hr at 30 cm from the source. At that time, the auxiliary building 649' elevation was controlled as a radiation area. Also, radiation protection (RP) staff had administratively required that radiation protection technicians monitor radiological conditions any time personnel were in this area. The licensee had a pre-established posting plan for the PCS chemical decontamination activity that included posting this location as a Locked High Radiation Area (LHRA). However, this posting plan was to be initiated prior to chemical injection, which had not occurred yet. Upon identification of the LHRA conditions, the licensee guarded the area and subsequently implemented the pre-established posting plan for all areas potentially affected by the PCS decontamination activities. This included posting and controlling this area as an LHRA. Corrective Actions: The licensee immediately guarded the area and subsequently posted and controlled the area as an LHRA.			

Corrective Action References: PAL-02617 and PAL-03310
<u>Performance Assessment:</u> None
<u>Enforcement:</u> Severity: The inspectors determined this was a SLIV NCV in accordance with the enforcement policy. Specifically, the failure to post and control the LHRA was of more than minor concern but resulted in relatively inappreciable potential safety consequences. Violation: TS 5.7.2 requires, in part, that entryways into High Radiation Areas (HRAs) with dose rates greater than 1.0 rem/hr at 30 centimeters from the radiation source be conspicuously posted as an HRA and be provided with a locked or continuously guarded door or gate that prevents unauthorized entry. Contrary to the above, on July 6, 2024, the licensee failed to conspicuously post the entryway to an area with dose rates greater than 1.0 rem/hr at 30 centimeters from the radiation source as an HRA and failed to lock or continuously guard the entryway to prevent unauthorized entry. Enforcement Action: This violation is being treated as a non-cited violation, consistent with Section 2.3.2 of the Enforcement Policy.

Observation: Chemical Decontamination Process of the Primary Coolant System (Occupational Radiation Safety Cornerstone)	71124.01
On July 7–10, 2024, Palisades performed a chemical decontamination process of the primary coolant system (PCS). The process involved the injection of mixed chemicals into the PCS to remove radiological contamination from piping to reduce worker dose for future activities. The inspectors observed the system setup and connections, initial radiation protection postings, initial chemical injection, and system cleanup. During the process, three of the unit's four primary coolant pumps were operated. Overall, the inspectors observed that the activity was done safely. They noted there were several personnel new to the station or new to their roles and saw some instances where communications regarding the activities could have been clearer. An example of this was associated with the initial radiation protection postings prior to the start of the chemical injection process. The RP field supervisor identified several postings that needed to be corrected after individuals initially hung the postings in the field. Although these postings were corrected prior to the start of the activity, it demonstrated unclear directions for the individuals tasked with hanging the initial radiation protection postings. It is important to ensure that work activities are appropriately understood, roles and responsibilities are clear, and, in this specific example, in-field radiation protection postings provide the correct instructions and barriers for workers to prevent occupational radiation safety issues.	

EXIT MEETINGS AND DEBRIEFS

The inspectors verified no proprietary information was retained or documented in this report.

- On November 7, 2024, the inspectors presented the inspection results to M. Mlynarek, Site Vice President, and other members of the licensee staff.

DOCUMENTS REVIEWED

Inspection Procedure	Type	Designation	Description or Title	Revision or Date
71111.08P	Corrective Action Documents Resulting from Inspection	PAL-02741	M&TE ID#016044 Used While Out of Calibration	07/25/2024
	Miscellaneous	51-9379921-000	Palisades Unit 1 MRP Bolting Ultrasonic Examination Plan	0
		INR-Palisades Restart-24-001	Remote Visual Examination Indication Notification Report	07/30/2024
		Visual Exam Plan	Palisades Reactor Vessel and Internals Remote Visual Examinations	07/23/2024
	NDE Reports	L-1 Crane Hooks	Magnetic Particle Examination Report	04/22/2024
		LD-1	Magnetic Particle Examination Report	04/22/2024
		LD-1	Ultrasonic Calibration and Exam Report	06/18/2024
		LD-1	Visual Examination Report	04/22/2024
	Procedures	CLP-M-6	Inspection of Heavy Load Lift Devices	08/22/2018
	Work Orders	50083288-01-001	L-1 Polar Crane PMs	02/06/2024
71111.11Q	Miscellaneous		Shift 3 Crew Notebook	09/17/2024
		PLSEG-LOR-24C-05	Palisades 2024 Cycle 24C Simulator Scenario	1
	Procedures	PNT-114	Licensed Operator Requalification Training Program Description	1
		PNT-210	Conduct of Simulator Training	0
		PNT-217	Examination Security	1
71111.18	Drawings	SK-C-417	Reactor Auxiliary Building Floor Plan EL.639'-0", 649'-0"	76
		WDD-DW-01160-CPAL1	Chemical Decontamination Equipment Layout Plan	4
71124.01	Miscellaneous		Radiological Survey Sheet for Auxiliary Building 649'	07/06/2024
			Primary Coolant System Decontamination Posting Plan	07/06/2024
		20240274	Radiation Work Permit and ALARA Plan for Primary Coolant System Full System Chemical Decontamination	01
		TSR-2024-155	Temporary Shielding Request for Water Shield Tanks Around Cask Washdown Pit	06/07/2024
		TSR-2024-157	Temporary Shielding Request for 590' Shield Around Floor	06/20/2024

Inspection Procedure	Type	Designation	Description or Title	Revision or Date
	Procedures		Plug	
		WI-RSD-R-008	Spent Resin Transfers to Resin Liners in Track Alley	25
		EN-RP-101	Access Control for Radiological Controlled Areas	0
		WDD-IP-01076-CAPL	Palisades 2024 FSD Installation Procedure	0
		WDD-OP-00085-CPAL1	Palisades 2024 FSD Operating Procedure	0